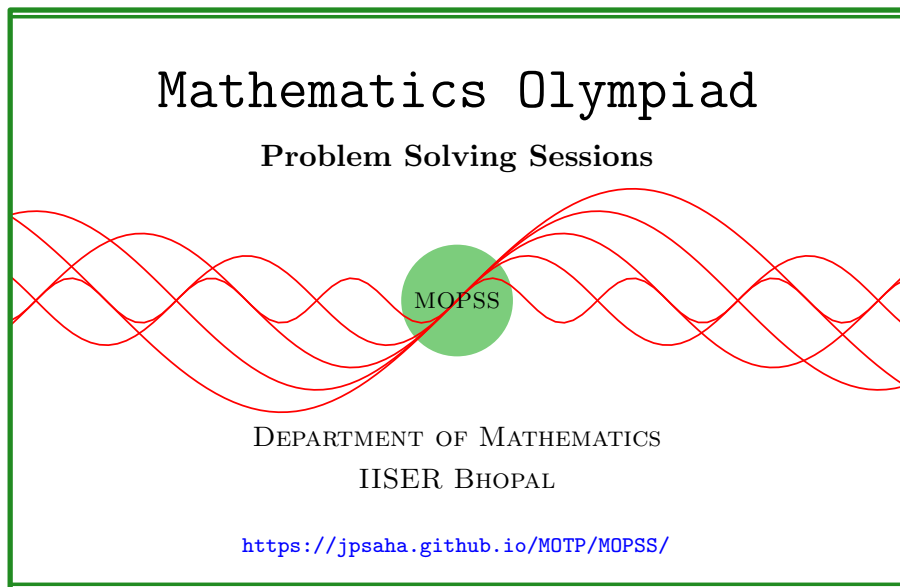


Generating functions

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

List of problems and examples

1.1 Example 2

§1 Generating functions

See [Sob13, Chapter 6], [Wil06].

Example 1.1. In how many ways, can we fill a bag with n fruits subject to the following constraints?

- 1. The number of apples must be even.
- 2. The number of bananas must be a multiple of 5.
- 3. There can be at most four oranges.
- 4. There can be at most one pear.

Solution 1. Note that the required number is equal to the coefficient of x^n in the formal power series $A(x)B(x)O(x)P(x)$ where

$$\begin{aligned} A(x) &= 1 + x^2 + x^4 + x^6 + \dots, \\ B(x) &= 1 + x^5 + x^{10} + x^{15} + \dots, \\ O(x) &= 1 + x + x^2 + x^3 + x^4, \\ P(x) &= 1 + x. \end{aligned}$$

Also note that

$$\begin{aligned} A(x)B(x)O(x)P(x) &= \frac{1}{1-x^2} \frac{1}{1-x^5} \frac{1-x^5}{1-x} \times (1+x) \\ &= \frac{1}{(1-x)^2} \\ &= 1 + 2x + 3x^2 + 4x^3 + \dots \end{aligned}$$

So the bag can be filled in $n + 1$ ways such that the given conditions hold. ■

References

[Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)

[Sob13] PABLO SOBERÓN. *Problem-solving methods in combinatorics*. An approach to olympiad problems. Birkhäuser/Springer Basel AG, Basel, 2013, pp. x+174. ISBN: 978-3-0348-0596-4; 978-3-0348-0597-1. DOI: [10.1007/978-3-0348-0597-1](https://doi.org/10.1007/978-3-0348-0597-1). URL: <http://dx.doi.org/10.1007/978-3-0348-0597-1> (cited p. 2)

- [**Wil06**] HERBERT S. WILF. *generatingfunctionology*. Third. A K Peters, Ltd., Wellesley, MA, 2006, pp. x+245. ISBN: 978-1-56881-279-3; 1-56881-279-5 (cited p. [2](#))